

Title:

SkillSync Hub: A Collaborative Learning and Skill Exchange Platform

Abstract:

The rapid growth of technology and the increasing demand for continuous learning have made skill development essential for students and professionals. However, many learners cannot access quality education due to expensive courses, while skilled individuals often lack a platform to share their expertise. Existing online learning platforms mainly focus on paid courses and provide limited opportunities for collaborative learning and skill exchange.

SkillSync Hub is a web-based Collaborative Learning and Skill Exchange Platform that enables users to learn and teach skills through multiple learning options. Users can exchange skills for free, enroll in free courses, or access premium courses with certificates upon successful completion. The platform supports technical, professional, creative, and personal development skills. Mentors can create courses, upload learning materials, schedule live sessions, interact with learners, and assess their progress. The system includes user authentication, skill matching, session scheduling, ratings and reviews, certificate generation, and an administrator dashboard.

Developed using HTML, CSS, JavaScript, Java Spring Boot, and MySQL, the platform provides a secure, scalable, and user-friendly environment. Future enhancements include AI-based personalized recommendations, multilingual support, mobile application development, and integrated video conferencing.

Existing System:

- Most online learning platforms provide paid courses and subscription-based learning.
- Peer-to-peer skill exchange is limited or unavailable.
- Free learning resources often do not provide structured guidance or certification.
- Learners need separate platforms for courses, live classes, communication, and certification.
- Limited interaction between learners and instructors.
- Personalized learning and collaborative knowledge sharing are limited.

Limitations of Existing System:

- High course and certification costs.
- Lack of free skill exchange opportunities.
- Limited collaboration among learners.

- No integrated platform combining skill exchange, courses, and live learning.
- Difficult to find suitable mentors or learning partners.
- Limited personalized learning experience.
- Reduced community engagement.
- Dependency on multiple applications for different learning activities.

Proposed System:

SkillSync Hub is a centralized web-based platform that combines collaborative learning, skill exchange, online courses, and mentorship within a single application. Users can register as learners or mentors, create profiles, and list the skills they can teach and the skills they wish to learn. The platform offers three learning options:

- **Free Skill Exchange** – Users exchange skills without any payment.
- **Free Courses** – Users can enroll in free courses created by mentors.
- **Premium Courses** – Users can access premium courses with live classes, assessments, and digital certificates.

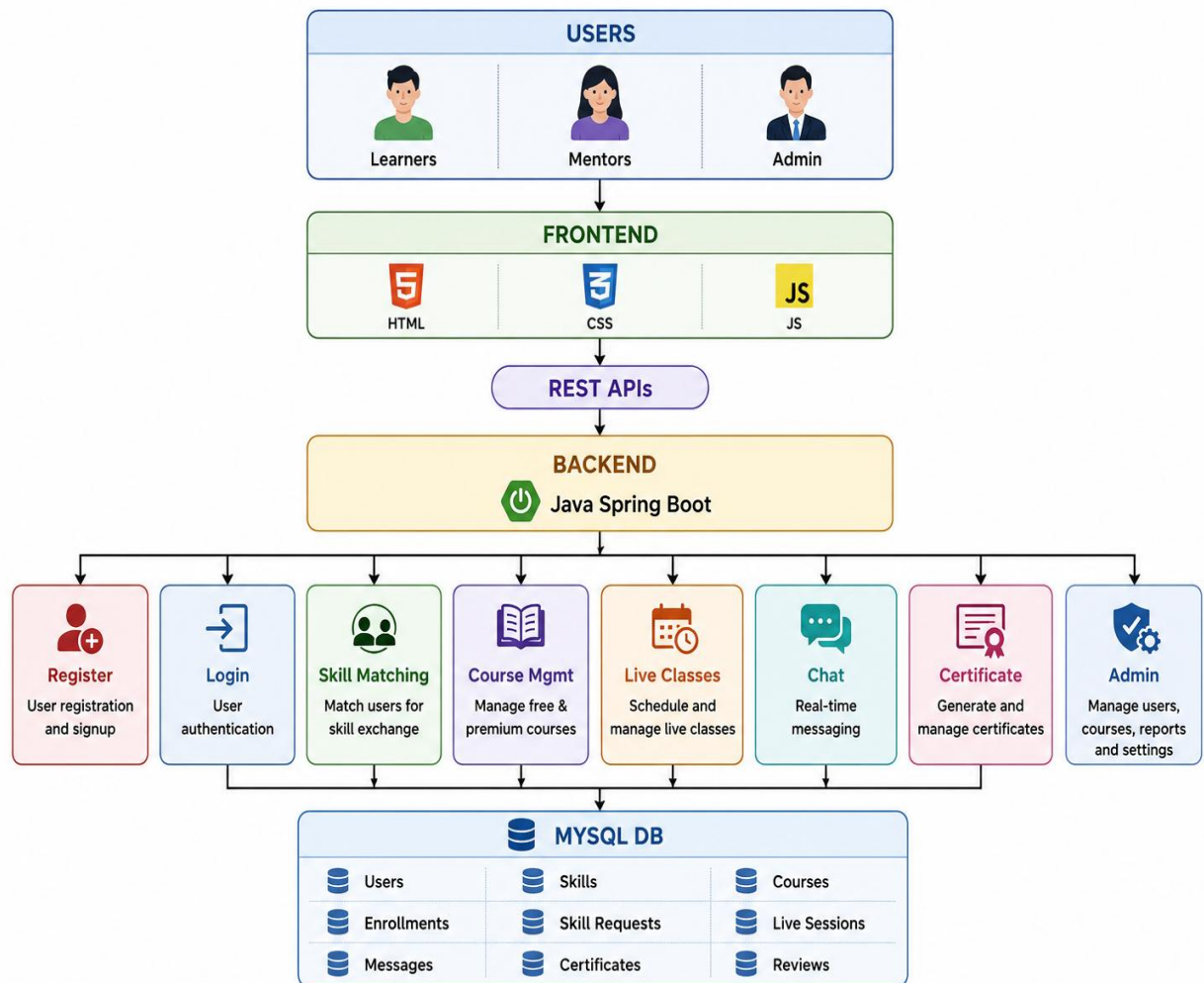
The system automatically matches users based on complementary skills, enables chat and discussion forums, schedules live learning sessions, tracks learner progress, manages certificates, and provides an administrator dashboard for platform monitoring and management.

Proposed System Advantages + Architecture

Advantages of Proposed System

- Provides free peer-to-peer skill exchange.
- Supports free and premium learning options.
- Offers live interactive classes.
- Generates digital certificates for premium courses.
- Encourages collaborative and community-based learning.
- Reduces the cost of quality education.
- Secure authentication and user management.
- User-friendly and scalable architecture.
- Supports technical, creative, and professional skills.
- Improves networking and career development opportunities.
- All learning activities are managed through a single platform

System Architecture



References

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